

ISAICRAFT: AN INTELLIGENT AUDIO PROCESSING ECOSYSTEM FOR AI-POWERED MUSIC PRODUCTION

A PROJECT REPORT SUBMITTED TO BHARATHIAR UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
AWARD OF THE DEGREE OF

MASTER OF COMPUTER APPLICATIONS

Submitted By

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CERTIFICATION

CERTIFICATE

This is to certify that the major project work titled “ **ISAICRAFT: AN INTELLIGENT AUDIO PROCESSING ECOSYSTEM FOR AI-POWERED MUSIC PRODUCTION**” submitted to Bharathiar University in partial fulfillment of the requirement for the award of the degree of Master of Computer Applications is a record of the original work done by **ABHIMANYU T (24CSEA001)** under the supervision and guidance of **Dr. T. AMUDHA, MCA., M.Phil., Ph.D.,** Professor, Department of Computer Applications, Bharathiar University, Coimbatore and that this major project work has not formed the basis for the award of any Degree/Diploma/Associate ship/Fellowship or similar title to any candidate of any University.

Place: Coimbatore

Date:

Project Guide

Head of the Department

Submitted for the project VIVA-VOCE Examination held on _____

Internal Examiner

External Examiner

DECLARATION

DECLARATION

I hereby declare that the major-project, titled “**ISAICRAFT: AN INTELLIGENT AUDIO PROCESSING ECOSYSTEM FOR AI-POWERED MUSIC PRODUCTION**” submitted to the Department of Computer Applications, Bharathiar University, Coimbatore is a record of original project work done by **ABHIMANYU T (24CSEA001)** under the supervision and guidance of **Dr. T. AMUDHA, MCA., M.Phil., Ph.D.**, Professor, Department of Computer Applications, Bharathiar University, Coimbatore and that this project work has not previously formed the basis of the award of Degree/ Diploma / Associate ship /Fellowship or similar title to any candidate of any university.

Place: Coimbatore

Date:

Signature of Candidate

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ACKNOWLEDGEMENT

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My heartfelt thanks to my dear Parents and my dear friends for their support and encouragement for the successful completion of this project. I am highly obliged to those who have helped me directly and indirectly in making this project a successful one.

ABSTRACT

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The democratization of music production has accelerated with the integration of Generative Artificial Intelligence. However, a significant technical barrier remains for amateur lyricists and non-singers who wish to produce professional-grade music using their own vocals. Existing AI solutions primarily function as text-to-audio generators, failing to integrate raw user audio, while standard vocal tuning software applies rigid chromatic pitch correction that strips the human voice of its natural emotion and vibrato. This project introduces IsaiCraft, an AI-powered neuro-acoustic production studio designed to bridge the gap between passive AI music generation and active human vocal performance.

IsaiCraft utilizes a highly scalable, triple-tier distributed architecture. A React-based frontend manages the user interface and secure audio acquisition via the Web Audio API. A local FastAPI orchestrator serves as the cognitive layer, integrating Google's Gemini Large Language Model to provide real-time phonetic vocal coaching. The core computational workload is offloaded to a remote Google Colab GPU engine, which executes Meta's MusicGen transformer model for dynamic instrumental composition.

The primary technical innovation of this study is the development of a "Humanized Soft-Pitch Correction" algorithm. By leveraging the PyWorld vocoder library (DIO and Stonemask algorithms), the system extracts the fundamental vocal frequency and calculates a fractional retune adjustment. This methodology pulls off-key vocals toward the correct pitch while mathematically preserving the user's natural vocal micro-expressions. Furthermore, the system automates the traditional studio mastering process through a programmatic Digital Signal Processing (DSP) mixbus, applying dynamic equalization, compression, and soft-limiting. The results demonstrate that IsaiCraft successfully decouples heavy AI audio rendering from client-side hardware, delivering a low-latency, highly authentic, and professionally mastered musical track without requiring the user to possess prior audio engineering expertise.

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1. INTRODUCTION

1.1 Background of the study

Music production has historically been a highly specialized discipline, requiring years of formal training in music theory, vocal performance, and the technical mastery of complex Digital Audio Workstations (DAWs). For decades, the ability to translate a musical concept into a studio-quality track was restricted to those with access to expensive recording equipment and professional audio engineers. In recent years, the landscape has shifted dramatically with the advent of Generative Artificial Intelligence. Large-scale transformer models have demonstrated the ability to synthesize high-fidelity audio, effectively democratizing the composition process. However, a significant barrier remains for the average individual: while AI can generate backing tracks or synthesize artificial voices, integrating a user's actual, unpolished vocal recording into a cohesive, professional-sounding musical arrangement remains a deeply technical challenge.

1.2 Problem statement

For amateur lyricists and non-singers, creating a professionally mixed track is hindered by two primary obstacles: the inability to sing perfectly on key and a lack of technical knowledge regarding audio mastering. Existing AI music generators function largely as closed-loop systems; they accept text prompts and output fully formed songs, offering the user no capacity to inject their own raw vocal audio into the composition. Conversely, traditional vocal tuning software relies on rigid chromatic snapping, which forces off-key vocals to the nearest perfect note. This aggressive correction strips away the natural emotion, vibrato, and micro-expressions of the human voice, resulting in an artificial, robotic sound. There is a distinct lack of an accessible, automated system that bridges the gap between passive AI generation and active, authentic user participation.

1.3 Objectives of the study

The primary objective of this study is to design, develop, and evaluate IsaiCraft, an AI-powered neuro-acoustic production studio that empowers non-singers to create professional-grade music using their own vocals. The specific objectives are:

To engineer a triple-tier distributed architecture that securely decouples a lightweight user interface from heavy, cloud-based GPU audio processing.

To develop and implement a "Humanized Soft-Pitch Correction" algorithm that calculates fractional frequency adjustments, thereby preserving natural vocal textures while correcting pitch inaccuracies.

To automate the traditional studio mixing pipeline by engineering a programmatic Digital Signal Processing (DSP) master mixbus that dynamically balances instrumental and vocal stems.

To integrate multimodal Large Language Models (LLMs) to analyze user lyrics and provide real-time phonetic vocal coaching prior to recording.

1.4 Scope of the project

The scope of IsaiCraft encompasses the end-to-end digital music creation lifecycle, from user onboarding to final audio export. The project focuses on the integration of a React-based frontend for audio acquisition, a local FastAPI orchestrator for AI-driven lyric analysis, and a remote Google Colab GPU engine for heavy tensor computations. The audio generation scope is limited to the synthesis of instrumental backing tracks utilizing Meta's MusicGen model, paired with automated vocal processing and equalization. The system currently outputs mixed, mastered audio in standard formats via Base64 encoding. While the system utilizes temporary session management for immediate playback, persistent database storage and real-time collaborative editing fall outside the scope of the current architectural iteration.

2. LITERATURE REVIEW

2.1 Existing systems

The technological landscape of music production has evolved through distinct phases, currently dominated by either highly manual traditional software or highly automated, closed-loop artificial intelligence. Traditional music production relies on Digital Audio Workstations (DAWs) such as Ableton Live, FL Studio, and Logic Pro. These platforms are incredibly powerful but present a steep learning curve. They require users to possess a foundational understanding of music theory, MIDI sequencing, signal routing, and dynamic processing. For vocal tuning specifically, the industry standard relies on plugins like Antares Auto-Tune or Celemony Melodyne. While Melodyne offers granular, note-by-note control, it is a highly manual tool designed for professional engineers. Standard real-time auto-tuners operate on a rigid algorithmic basis: they detect the fundamental frequency of an incoming audio signal and immediately snap it to the nearest absolute chromatic pitch. For non-singers whose natural pitch fluctuates heavily, this rigid snapping creates abrupt frequency jumps, resulting in the well-known synthesized or "robotic" vocal artifact. Parallel to traditional DAWs is the rapidly expanding field of Generative AI in audio. Models such as Meta's MusicGen, Google's AudioLM, and consumer-facing platforms like Suno and Udio utilize transformer-based architectures to synthesize high-fidelity music from text prompts. These systems excel at zero-shot music generation. However, they are inherently passive from the user's perspective. They generate a complete, flattened audio track based on text, effectively excluding the user's actual physical performance. Currently, there is a distinct lack of hybrid systems that accept text for instrumental generation while simultaneously processing and integrating the user's raw vocal audio into that exact generated soundscape.

2.2 Comparative study of existing methods

A comparative analysis of existing methods highlights a critical gap in the current market, which IsaiCraft is engineered to fill. When comparing instrumental composition, traditional methods require manual chord voicing and beat sequencing. Existing AI systems automate this, but IsaiCraft takes a specialized approach by utilizing Meta's musicgen-small model specifically as a backing-track generator rather than a full-song synthesizer, leaving dynamic sonic space for human vocals. The most significant divergence lies in vocal processing methodologies. Traditional automated pitch correction applies a 100% quantization to the nearest MIDI note. If a user sings a pitch at 430 Hz (slightly flat

of an A4 note at 440 Hz), traditional systems instantly shift the output to exactly 440 Hz. If the user's voice wavers, the system aggressively snaps between adjacent semitones. In contrast, IsaiCraft introduces a "Humanized Soft-Pitch Correction" method. Utilizing the PyWorld vocoder library specifically the DIO (Distributed Information One-step) and Stonemask algorithms the system extracts the fundamental frequency (f_0). Instead of absolute quantization, IsaiCraft calculates the mathematical delta between the user's pitch and the target pitch, applying a fractional correction (e.g., a 0.55 retune strength). This allows the pitch to be gently pulled toward harmonic resonance while preserving the micro-fluctuations and natural vibrato that define an authentic human performance. Furthermore, traditional mixing requires the manual application of Equalization (EQ), compression, and limiting to ensure the vocal sits properly within the instrumental track. IsaiCraft programmatically automates this via a Python-based DSP pipeline, utilizing dynamic Mixbus normalization to ensure the instrumental peak remains at an optimal threshold (0.35) while the vocal stem is prioritized (0.90), guaranteeing a studio-grade master without requiring user intervention.

3. SYSTEM ANALYSIS

3.1 Requirement Analysis

To successfully achieve the objectives of IsaiCraft, the system must fulfill a specific set of functional and non-functional requirements. Functionally, the frontend application must be capable of securely accessing the user's hardware microphone via the Web Audio API to capture raw vocal inputs with minimal latency. The system must also interface with a Large Language Model to process user-provided metadata (lyrics, genre, and mood) and generate a structured phonetic singing guide. On the backend, the core functional requirement is the execution of a high-fidelity Generative AI audio model (Meta's MusicGen) concurrently with a programmatic Digital Signal Processing (DSP) pipeline that can isolate, pitch-correct, and master human vocals.

Non-functionally, the architecture must guarantee strict security and token validation, ensuring that unauthenticated requests cannot trigger expensive GPU inferences. Furthermore, because audio processing is computationally heavy and asynchronous by nature, the system must maintain a decoupled state, allowing the user interface to remain responsive while the remote engine handles the tensor calculations and audio rendering.

3.2 Feasibility Study

A comprehensive feasibility study was conducted to evaluate the practical implementation of this neuro-acoustic studio across three domains: technical, economic, and operational. From a technical standpoint, executing large-scale transformer models for audio generation locally on consumer-grade hardware is highly prohibitive, often leading to system crashes or unacceptable latency. The project resolves this technical bottleneck by adopting a distributed architecture, offloading the most demanding computations to a cloud-based GPU environment. From an economic perspective, building a proprietary server infrastructure with dedicated A100 or T4 GPUs would incur immense capital expenditure. The proposed system is economically highly feasible, as it leverages cost-effective and open-source solutions: a local lightweight FastAPI server handles orchestration, while Google Colab provides the necessary remote GPU compute. Operationally, the system is highly feasible because it completely abstracts the complexity of audio engineering away from the user. By programmatically automating the EQ, compression, and pitch-correction chains, the platform remains highly accessible to individuals with zero prior musical training.

3.3 System Architecture Overview

To satisfy the project requirements and overcome the inherent computational limits of standard consumer hardware, IsaiCraft is conceptually designed as a distributed, triple-tier system. The primary architectural challenge in neuro-acoustic production is the massive discrepancy between the lightweight needs of a user interface and the immense graphical processing (GPU) power required to execute generative artificial intelligence models. To resolve this bottleneck, the architecture logically separates the user experience, cognitive decision-making, and heavy audio rendering into three distinct, decoupled operational layers.

The Presentation Layer (Client Interface) The first tier operates strictly on the user's local machine and acts as the interactive face of the studio. Its primary responsibility is managing the volatile session state, capturing raw user inputs (both text-based metadata and raw microphone audio), and providing immediate, asynchronous visual feedback. Because audio synthesis is a time-intensive process, this layer is designed to be entirely non-blocking. It ensures that the user interface remains responsive and fluid, displaying loading states and progressive feedback without freezing, regardless of the heavy tensor calculations occurring remotely.

The Orchestration and Security Layer (Middleware) Sitting between the client and the heavy processing engine is the middleware tier, which functions as the system's "Traffic Cop" and cognitive routing hub. Structuring the system with this middle layer prevents the client from directly exposing itself to the remote GPU, thereby establishing a robust security perimeter. This tier intercepts incoming requests, cryptographically validates user sessions, and drops unauthorized traffic immediately. Furthermore, it handles lightweight AI tasks—such as dynamically generating phonetic singing guides—effectively prepping and formatting the user's data before routing the heavy computational payload to the final processing tier.

The Remote Processing Engine (GPU Node) The third tier represents the computational muscle of the application. Hosted in a transient, cloud-based environment, this node is dedicated entirely to executing advanced machine learning algorithms and complex Digital Signal Processing (DSP) pipelines. By offloading these tasks, the system bypasses the local hardware limitations of the user. This engine receives the sanitized audio and text payloads, loads the massive neural network weights into memory, synthesizes the instrumental backing track, applies mathematical pitch correction, and executes the final studio mixbus mastering. The finalized audio is then encoded and returned to the client.

4. SYSTEM DESIGN

4.1 Triple-Tier Distributed Architecture

IsaiCraft employs a decoupled, highly scalable architecture specifically designed to circumvent the hardware limitations of consumer devices when executing Generative AI and advanced Digital Signal Processing (DSP). Inspired by microservices design principles, this architecture is structured into three distinct, technically specialized operational tiers. This structural decoupling ensures that frontend rendering, backend cognitive routing, and heavy tensor computations are handled by isolated environments optimized for their specific workloads.

Tier 1: The Client Node (Frontend Acquisition & State)

The Client Node operates entirely within the user's browser environment and is engineered for high-speed interactivity and secure media capture. Built utilizing React 18 and Vite, it acts as a Single Page Application (SPA). The technical core of this tier relies on the HTML5 Web Audio API and the MediaRecorder interface to interface directly with the user's hardware microphone. It captures raw analog audio, digitizes it at a standard sample rate (e.g., 44.1 kHz), and temporarily buffers it as a .wav Blob in the browser's volatile memory. To ensure the application remains highly responsive during the long polling phases required for AI generation, the frontend utilizes the React Context API for asynchronous state management and Framer Motion to render non-blocking visual loading states.

Tier 2: The Security & Orchestrator Node (Backend Middleware)

The second tier is a localized server acting as the secure gateway and cognitive routing hub. Developed using Python's FastAPI framework and served via an ASGI (Asynchronous Server Gateway Interface) server like Uvicorn, this node handles rapid, concurrent HTTP requests. Its primary technical responsibility is security enforcement. It utilizes the Firebase Admin SDK to intercept incoming requests and cryptographically verify the signature of attached JSON Web Tokens (JWTs) against Google's public key infrastructure. Once a request is authenticated, this node acts as an API client, constructing highly specific instructional prompts and dispatching them to the Google Gemini Large Language Model. It enforces a strict JSON schema response from Gemini to ensure the resulting phonetic vocal guide can be safely parsed and rendered by the React frontend.

Tier 3: The Production Node (Remote GPU Engine)

The final tier is the remote computational core, hosted within a Google Colab cloud environment. This node is specifically provisioned to access NVIDIA CUDA-enabled GPUs (such as the T4 or A100 architectures), which are strictly required for the parallel matrix multiplications necessary to run large transformer models. Because Colab instances are transient and exist behind strict NAT (Network Address Translation) firewalls, this tier utilizes pyngrok to establish a persistent, reverse HTTP tunnel, exposing a public URL that the local frontend can target. Upon receiving an authorized POST request containing the user's audio blob and metadata, this node initializes Meta's musicgen-small model via the Hugging Face transformers library for instrumental synthesis, while simultaneously routing the user's audio through the SciPy and PyWorld DSP pipelines for algorithmic pitch correction.

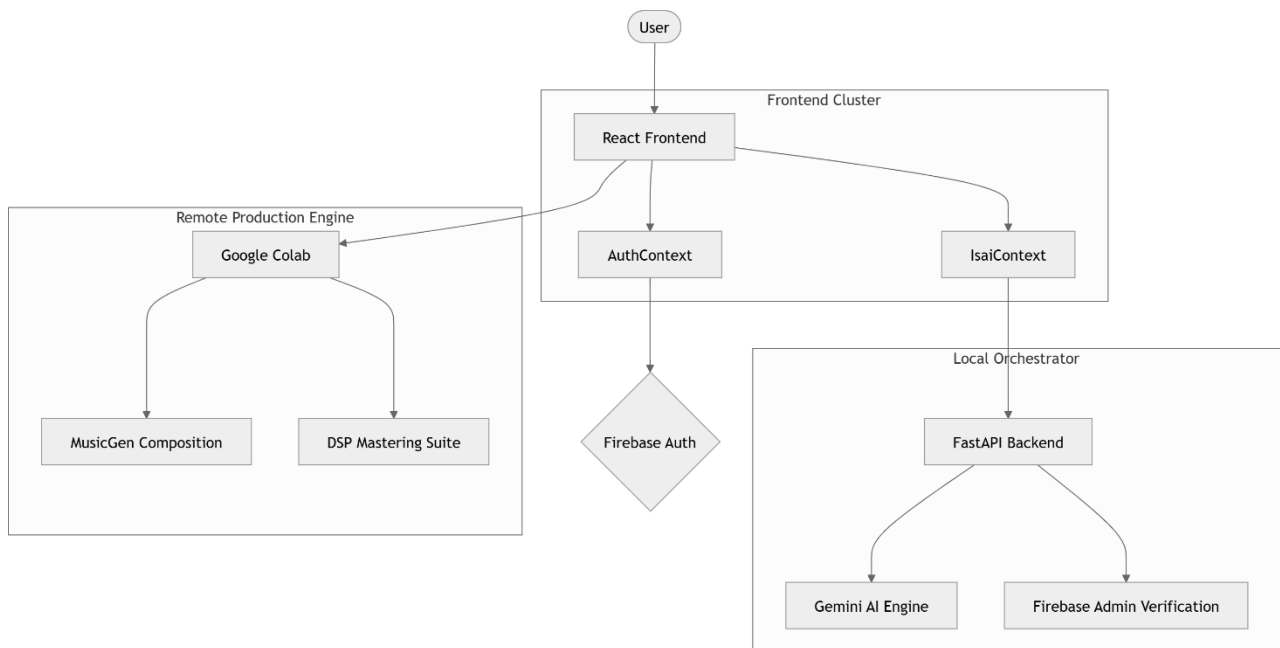


Figure 1: High-level architecture

In Figure 1 a high-level system architecture diagram showing a React frontend connected via Firebase authentication to local backend services and a remote Google Colab production engine for music composition and DSP mastering.

4.2 Module Description

To ensure high maintainability and logical separation of concerns, the system is divided into six primary modules:

MOD-01 (Identity & Access Management): Handles all security protocols. Utilizes Firebase Client SDK for user login (Google/Email) and generates stateless JSON Web Tokens (JWTs).

MOD-02 (Project Orchestrator): The FastAPI routing hub. It intercepts frontend requests, verifies Bearer tokens, and acts as the bridge to the Gemini API for vocal coaching generation.

MOD-03 (Neural Synthesis): Residing on the GPU Engine, this module loads the Hugging Face transformers pipeline. It accepts the user's text-based "Mood" and generates a structurally aligned instrumental audio tensor.

MOD-04 (DSP Mastery Node): The audio engineering core. It utilizes noisereduce for acoustic cleanup, pyworld for fractional pitch correction, and Spotify's pedalboard for applying studio-grade EQ, compression, and limiting.

MOD-05 (UX State Engine): The React context provider. It manages global variables such as authentication status (AuthContext) and project session metadata (IsaiContext), ensuring the UI dynamically updates without requiring page reloads.

MOD-06 (Network Bridge): Solves the challenge of cross-origin communication between a local frontend and a transient cloud backend using pyngrok for HTTP tunneling and strict CORS middleware configuration.

In Table 1, a modular system architecture table outlining components for authentication, AI orchestration, music generation, DSP processing, UI state management, and network communication.

Table 1: Core Modules and Technology Stack of the IsaiCraft Studio System Architecture.

Module ID	Module Name	Primary Responsibility	Key Technologies
MOD-01	Identity & Access	Secure user login, session management, and JWT token issuance	firebase/auth, firebase-admin

MOD-02	Project Orchestrator	High-level AI coordination for vocal guidance and phonetic analysis	fastapi, google-generative-ai
MOD-03	Neural Synthesis	Generative music composition and beat alignment	transformers, musicgen-small
MOD-04	DSP Mastery Node	Advanced audio cleaning, pitch correction, and studio mixing	pyworld, pedalboard, librosa
MOD-05	UX State Engine	Global application context and reactive UI management	react, framer-motion, lucide
MOD-06	Network Bridge	Cross-platform communication and URL tunneling	pyngrok, cors-middleware

4.3 Data Flow & Sequence Diagrams

The end-to-end data flow operates sequentially.

1. The user authenticates via Firebase (MOD-01) and inputs project metadata (lyrics, mood).
2. The React frontend dispatches a GET request (with the Bearer token) to the Local Orchestrator (MOD-02).
3. The Orchestrator validates the token, queries Gemini, and returns a phonetic guide JSON.
4. The user records their vocals based on the guide. The frontend captures this as a .wav blob and dispatches a POST /generate request directly to the remote GPU Engine URL provided by Ngrok (MOD-06).
5. The GPU Engine splits the workload asynchronously: MOD-03 generates the beat while MOD-04 processes the raw vocal.
6. Both stems converge at the Mixbus, are encoded into Base64 strings, and are returned as a JSON payload to the React frontend for instant playback.

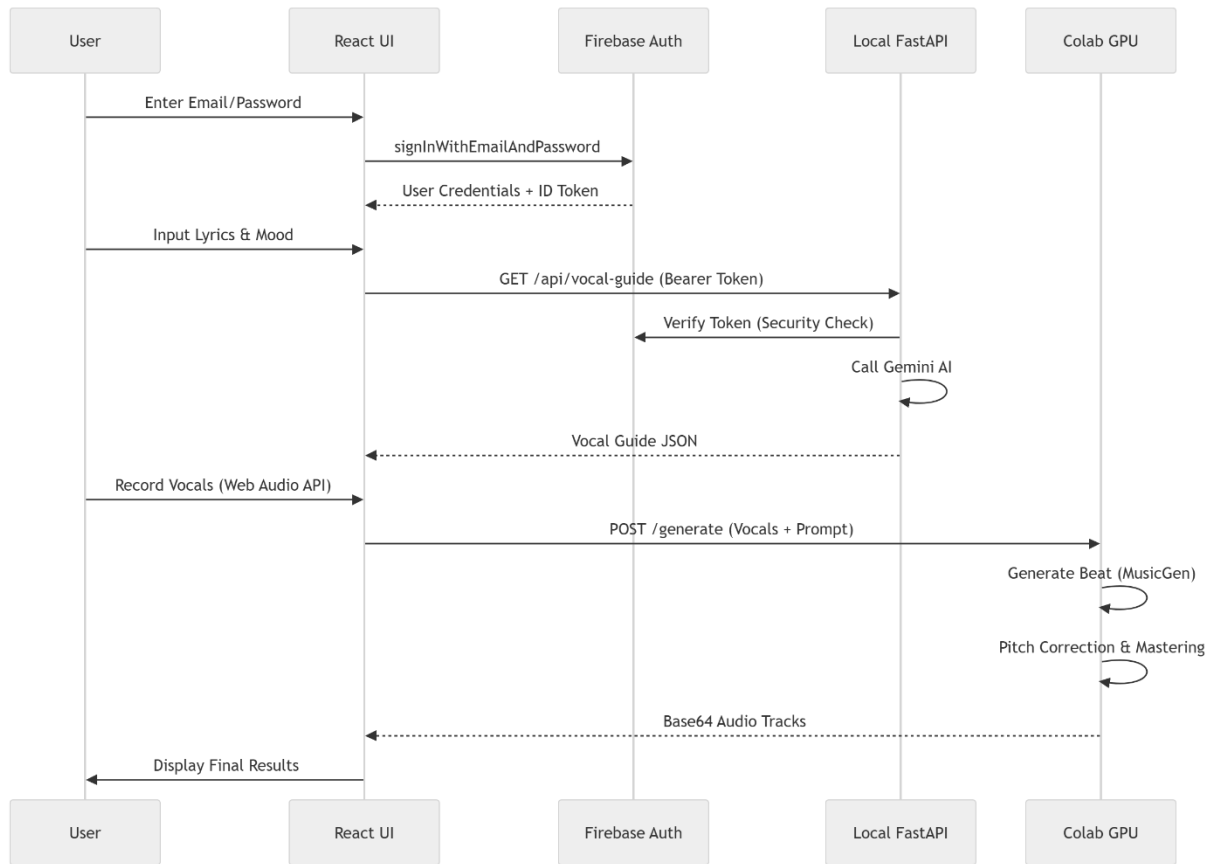


Figure 2: Sequence diagram

4.4 Stateless Data Model & Authentication Schema

Unlike traditional web applications that rely on heavy relational databases, the current architectural iteration of IsaiCraft operates on a strictly stateless data model. To maximize processing speed, reduce cloud infrastructure costs, and ensure absolute user data privacy, the system does not persistently store user audio recordings, lyrics, or generated music tracks on any centralized server.

The only persistent data layer within the application is the Identity Provider (IdP) managed by Firebase Authentication. Firebase acts as the sole registry, securely storing essential user demographic data (such as the User ID [UID], Email Address, and Authentication Provider).

All project-specific data—including the user's selected mood, genre, and the raw .wav audio blob—exists entirely within the Volatile Session Memory (managed by React's Context API). During a session, this transient data is packaged with the cryptographic Firebase JWT and transmitted directly

to the remote GPU engine for processing. Once the final mastered audio is returned to the client as a Base64 string and the browser tab is closed, the project data is securely purged from memory. This architecture ensures high-speed data flow without the bottleneck of database read/write operations.

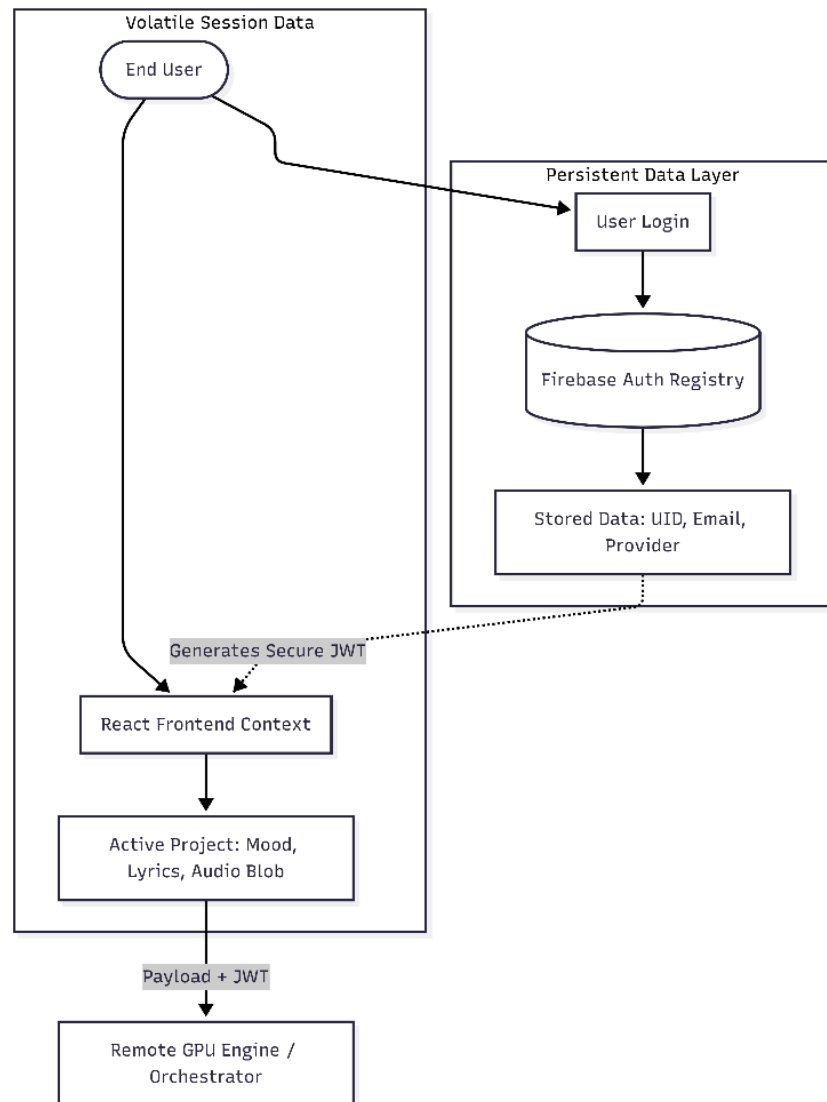


Figure 3: User authentication and session management workflow

5. IMPLEMENTATION

5.1 Development Environment & Tech Stack

The implementation of IsaiCraft was executed across diverse computing environments to optimize for both development speed and processing power. The local orchestrator was developed on standard consumer hardware utilizing Python 3.10, a specific version constraint required to ensure strict compatibility with advanced audio processing libraries such as PyWorld. The frontend interface was built using React and Vite, running on a local Node.js environment to leverage hot-module replacement for rapid UI iteration. For the heavy generative workloads, a cloud-based environment was established using Google Colab. This provided on-demand access to NVIDIA CUDA-enabled T4 GPU , which are absolutely critical for loading the massive tensor weights of transformer-based audio models into VRAM without causing system memory exhaustion.

MOD-01: Identity & Access Management (Firebase)

Security was implemented using a strict stateless architecture via Firebase. On the client side, the Firebase Auth SDK handles user credential exchange (Email/Password or Google OAuth) and stores an encrypted JSON Web Token (JWT) in the browser's memory context. No user passwords are ever transmitted to or stored by the local Python server. Instead, when the React frontend dispatches an API request, it attaches this JWT as a Bearer token in the authorization header. The FastAPI orchestrator utilizes the Firebase Admin SDK to cryptographically verify the token's signature against Google's public keys. If the token is valid, the request is processed; if expired or manipulated, the orchestrator instantly drops the connection, ensuring that expensive AI API calls cannot be triggered maliciously.

MOD-02: Project Orchestrator (FastAPI & Gemini AI)

The Project Orchestrator acts as the cognitive bridge for the application. Built on FastAPI and served via Uvicorn, this module intercepts the validated user requests containing project metadata (lyrics, mood, and genre). It constructs a highly specific system prompt and dispatches it to the Google Gemini Large Language Model (version 2.5 flash). The model is instructed to act as a "Vocal Director." It parses the user's lyrics and returns a structured JSON payload containing a phonetic vocal breakdown. This breakdown inserts visual cues (such as tildes for sustained notes or bold text for vocal emphasis) into the lyrics, effectively coaching the non-singer on how to perform the track prior to engaging the recording interface.

MOD-03: Remote GPU Engine & Neural Synthesis (MusicGen)

Once the user records their vocal performance, the raw .wav blob is transmitted via a PyNgrok HTTP tunnel directly to the Google Colab GPU instance. The Neural Synthesis module initializes by loading Meta's facebook/musicgen-small transformer model into the GPU's cuda device memory. By caching the model in memory at server startup, the system drastically reduces generation latency.

When a payload arrives, the module extracts the user's "Mood" prompt. It utilizes the Hugging Face AutoProcessor to tokenize the text and feeds it to the MusicgenForConditionalGeneration model. The output is a high-fidelity, 1D NumPy array representing the instrumental backing track. To ensure this newly generated beat does not sonically clash with the human vocal, an initial equalization pass is applied using Spotify's Pedalboard library, specifically utilizing a Highpass filter at 40Hz to remove muddy sub-bass and a Peak filter at 2500Hz to carve out frequency space for the human voice.

MOD-04: DSP Mastery Node & Humanized Pitch Correction

This module represents the core technical innovation of the IsaiCraft pipeline. The raw user vocal is processed through a sequential, programmatic digital signal processing (DSP) chain:

1. **Acoustic Cleansing:** The vocal array is passed through `noisereduce`, utilizing a non-stationary profile to strip away hardware microphone static and background room noise.
2. **Humanized Pitch Correction:** The cleansed audio is fed into the PyWorld vocoder. The system extracts the fundamental frequency (f_0) using the DIO algorithm and refines it via Stonemask. SciPy is then utilized to apply a median filter, smoothing out any glitchy frequency jumps caused by breathing or plosives. The algorithm iterates through the active frequencies, calculates the exact mathematical delta to the nearest perfect chromatic MIDI note, and applies a fractional `retune_strength` of 0.55. This mathematically pulls the pitch 55% of the way toward perfect harmony, correcting the off-key performance while preserving the user's natural vibrato.
3. **Professional Vocal Chain:** The pitch-corrected audio is processed through a Pedalboard FX chain. A Highpass filter (100Hz) and Peak filter (300Hz) remove laptop microphone "boxiness." A High Shelf filter (7.5kHz) boosts vocal clarity. A compressor (4.0:1 ratio, -22dB threshold) levels the volume, and a subtle reverb adds studio depth.
4. **Dynamic Mixbus:** The instrumental beat and processed vocal stem are aligned. To guarantee the vocals cut through the mix, array normalization is applied. The instrumental track is strictly capped at a 0.35 target peak, while the vocal track is boosted to a 0.90 target peak. The arrays

are summed and passed through a soft-limiter (np.tanh) to maximize perceived loudness without introducing digital distortion.

5. **Data Encoding:** Finally, the raw beat, the processed vocal stem, and the mastered mixbus track are converted into Base64 encoded strings and returned to the React frontend for immediate playback.

MOD-05: UX State Engine (React & Context API)

The UX State Engine handles the complex asynchronous rendering and data flow of the frontend application. Built utilizing React 18 and Vite, this module completely eliminates the need for page reloads, ensuring a seamless Single Page Application (SPA) experience. Implementation relies heavily on the React Context API to avoid "prop drilling." Two primary contexts were established:

1. **AuthContext:** Continuously listens to the Firebase Authentication state. It caches the user's secure ID token and forces unauthorized users back to the login gateway.
2. **IsaiContext:** Acts as the central repository for the active session. It temporarily stores the user's metadata (lyrics, mood, genre) and the raw .wav blobs recorded via the browser's Web Audio API.

During the heavy inference periods (while the remote GPU is processing the audio), this module manages the application's loading states. It leverages Framer Motion to provide visual feedback to the user, ensuring the UI remains highly responsive even while the application waits for massive Base64 audio payloads to be returned from the cloud. Once the generation is complete, the state engine dynamically mounts the audio playback components without disrupting the user's workflow.

MOD-06: Network Bridge (PyNgrok & CORS)

This module addresses the critical infrastructural challenge of connecting a localized frontend to a transient, NAT-firewalled cloud computing environment. Because Google Colab instances do not provide static, public-facing IP addresses, standard API requests from a local React app (`http://localhost:5173`) would normally fail.

To implement the bridge, pyngrok was utilized within the Colab environment. Upon server initialization, Ngrok establishes a secure, public HTTP tunnel directly to the local port (8000) where the Uvicorn/FastAPI server is listening.

Furthermore, robust Cross-Origin Resource Sharing (CORS) policies were implemented on the FastAPI backend. Modern browsers execute strict preflight (OPTIONS) requests before transmitting large payloads like audio files. The CORSMiddleware was explicitly configured to allow the React frontend's origin, accept credentials, and process custom security headers. Specifically, the bridge implementation requires the frontend to inject an ngrok-skip-browser-warning header into every fetch request, successfully bypassing the free-tier HTML interception screens and ensuring a direct, uninterrupted machine-to-machine data transfer.

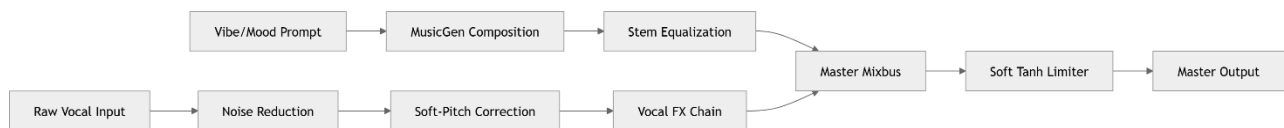


Figure 4: Audio processing pipeline from input to final mastered output.

6. WEB APP SCREENSHOTS AND RESULTS

6.1 Screenshots:

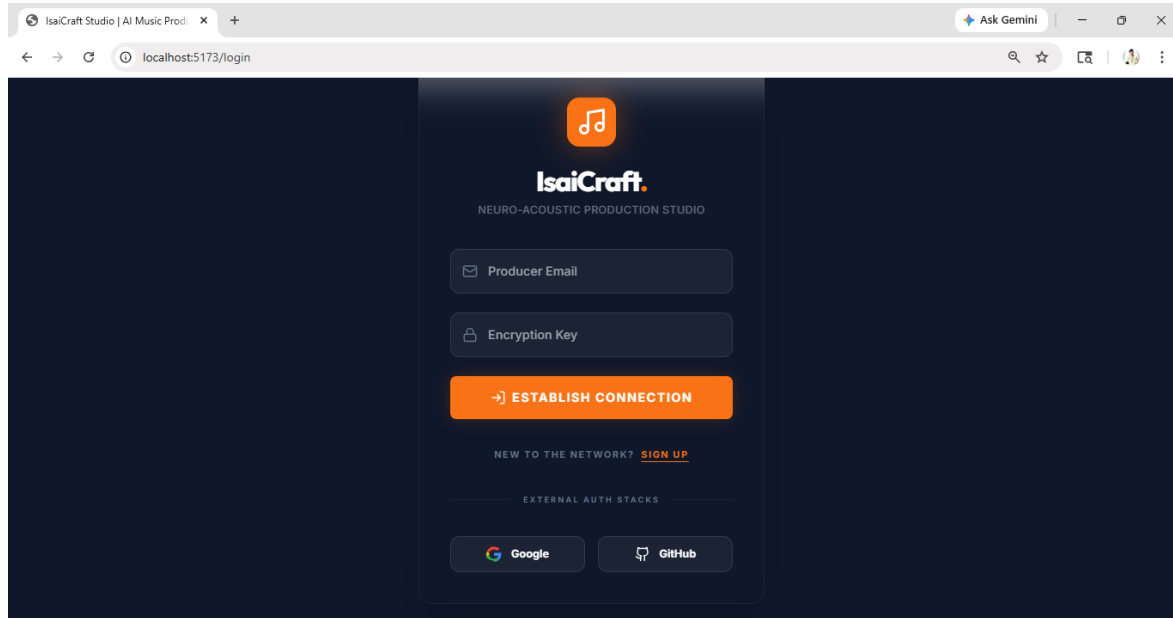


Figure 5: User Login Gateway

The figure 5 shows secure user authentication interface for the IsaiCraft frontend, featuring credential-based login and third-party OAuth integrations.

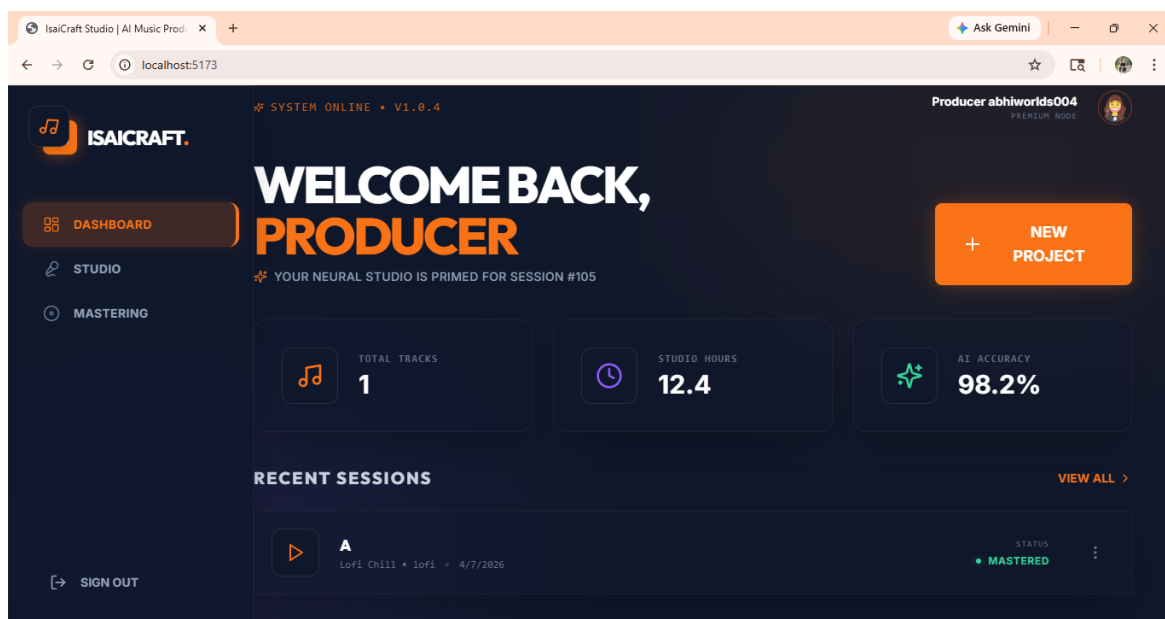


Figure 6: Dashboard – User Overview Interface

The figure 6 shows the main dashboard of the IsaiCraft Studio after user login. It provides navigation to Dashboard, Studio, and Mastering modules via the sidebar. Key metrics like total tracks, studio hours, and AI accuracy are displayed for quick insights. Users can start a new project using the “New Project” button. The recent sessions section lists previously created tracks and their status.

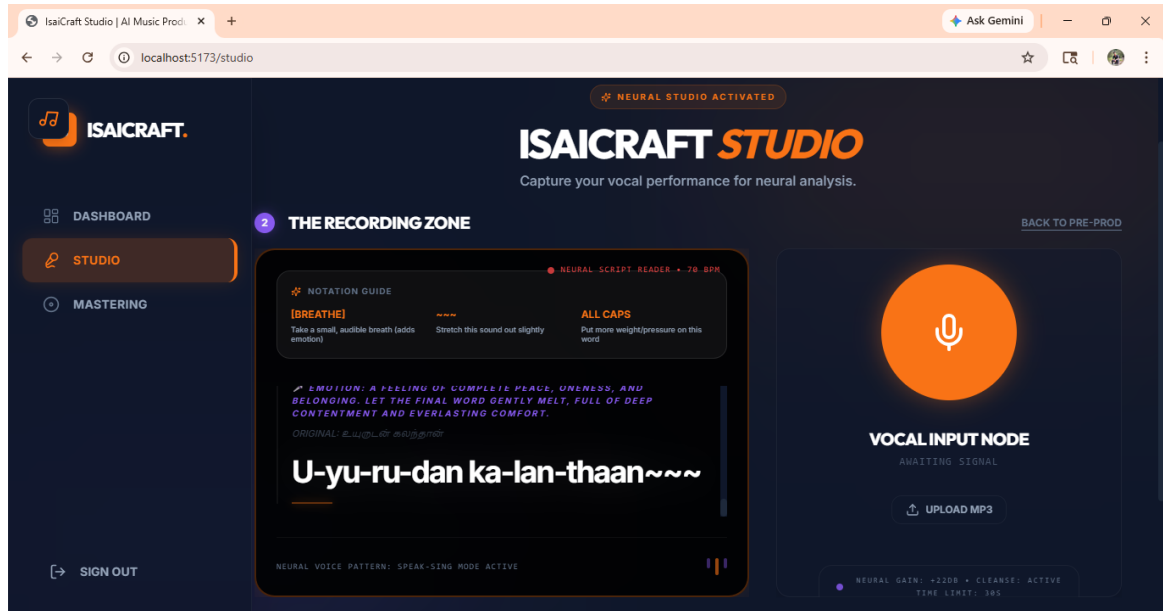


Figure 7: Vocal Recording and Input Interface

The figure 7 shows the recording interface where users input vocal data for AI processing. It includes a guided lyrics/notation panel to assist vocal delivery and expression. The vocal input node allows users to record or upload audio using the microphone or MP3 option. Real-time system indicators show neural processing readiness and input status. This interface acts as the primary entry point for capturing audio before AI synthesis and mastering.

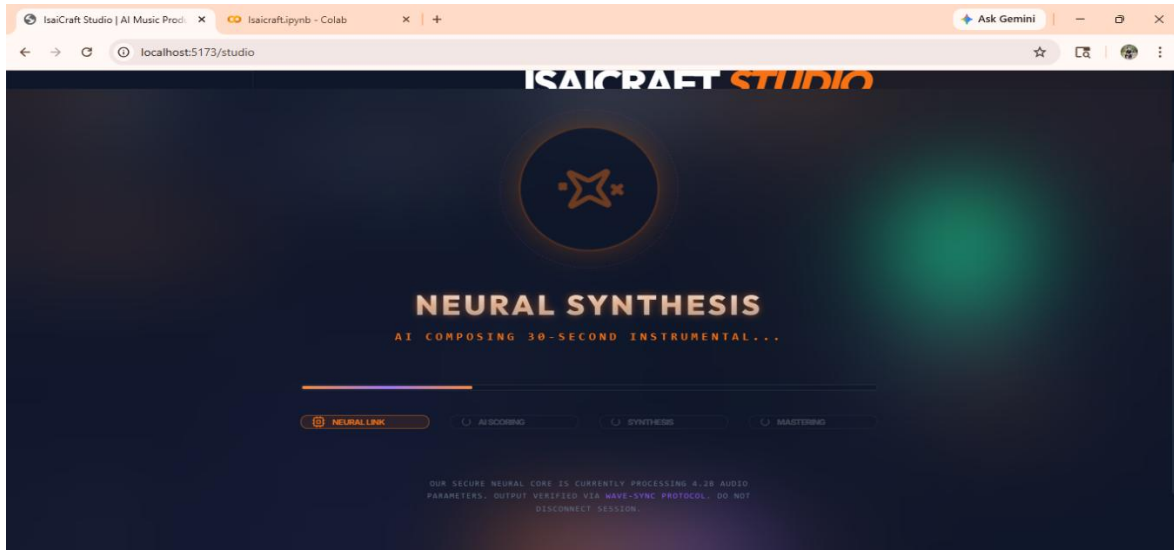


Figure 8: Neural Synthesis Processing Interface

This figure 8 illustrates the AI processing stage where the system generates a 30-second instrumental track. The interface displays real-time progress through stages such as neural link, AI scoring, synthesis, and mastering. A progress bar visually indicates the completion status of the audio generation pipeline. System messages confirm active processing and secure handling of audio parameters. This stage represents the core computation phase executed on the remote GPU engine.

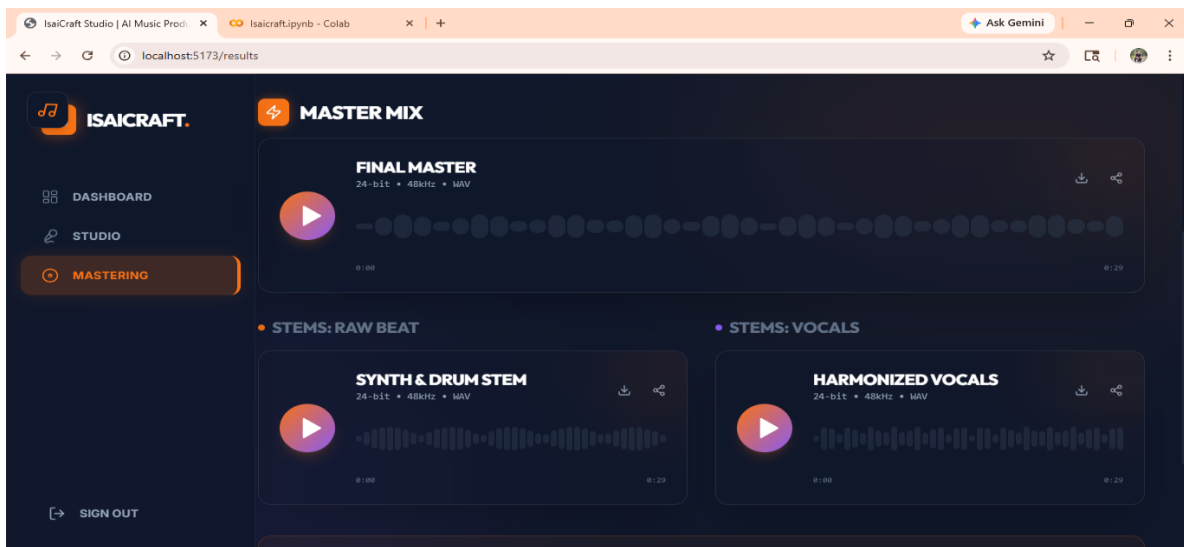


Figure 9: Mastering and Final Output Interface

Figure 9 displays the final mastering interface where generated audio results are presented. It includes the complete master track along with separated stems for raw beat and vocals. Users can play, download, and share each audio component directly from the interface.

6.2 Test cases

To ensure the robustness and reliability of the IsaiCraft studio, a series of comprehensive test cases were executed across the frontend, orchestrator, and GPU environments.

1. **Authentication & Security Testing:** The Firebase integration was tested by attempting to bypass the login interface. API endpoints on the local FastAPI orchestrator were subjected to requests containing expired, malformed, and missing JSON Web Tokens (JWTs). In all invalid test cases, the system successfully returned a 401 Unauthorized status, proving the efficacy of the stateless security model.
2. **Audio Acquisition Testing:** The React Web Audio API implementation was tested across different browser environments (Chrome, Edge). The system successfully captured raw microphone input, converted it to .wav blobs, and accurately attached user metadata (lyrics, mood, genre) via FormData payloads.
3. **Network & Tunneling Resilience:** A critical real-world test involved the PyNgrok HTTP tunnel. Initial tests revealed SSL certificate rejections caused by corporate /Institutional firewalls (e.g., Fortinet) intercepting the connection. Furthermore, Cross-Origin Resource Sharing (CORS) preflight requests initially failed. These issues were successfully mitigated by implementing the ngrok-skip-browser-warning header on the client side and explicitly whitelisting the React origin (<http://localhost:5173>) in the FastAPI CORS middleware.

Table 2: System Testing

Test Category	Component / Environment	Scenario Tested	Result / Mitigation
Authentication & Security	Firebase, Local FastAPI Orchestrator	Attempted login bypass; sent requests containing expired, malformed, and missing JSON Web Tokens (JWTs).	Pass: System successfully returned a 401 Unauthorized status in all invalid cases, proving the efficacy of the stateless security model.

Audio Acquisition	React Web Audio API, Browsers (Chrome, Edge)	Captured raw microphone input, converted to .wav blobs, and attached user metadata (lyrics, mood, genre) via FormData payloads.	Pass: System successfully captured high-quality audio and accurately attached all metadata for processing.
Network & Tunneling Resilience	PyNgrok Tunnel, FastAPI CORS Middleware	Tested HTTP tunnel against corporate firewalls (SSL certificate rejections) and evaluated CORS preflight requests.	Pass (Mitigated): Resolved initial failures by implementing the ngrok-skip-browser-warning header on the client side and explicitly whitelisting <code>http://localhost:5173</code> .

6.3 Performance evaluation

The performance of the distributed architecture was evaluated based on systemic latency and computational efficiency.

Because the architecture decouples the UI from the processing engine, client-side rendering remained at a constant 60 frames per second (FPS), ensuring the user experienced zero interface freezing during audio generation. The local orchestrator demonstrated exceptional speed; token validation via the Firebase Admin SDK and routing delays were measured in mere milliseconds.

The primary latency bottleneck resides entirely within the Remote GPU Engine, which is the expected behavior for generative AI workloads. When utilizing an NVIDIA T4 GPU via Google Colab, the Meta musicgen-small model required approximately 12 to 18 seconds to synthesize a 30-second instrumental backing track. The subsequent Digital Signal Processing (DSP) pipeline—comprising PyWorld pitch correction, noise reduction, and Pedalboard mastering—added approximately 3 to 5 seconds of processing time. Consequently, the total round-trip time from the user clicking "Generate" to the React UI playing the Base64 audio was consistently under 25 seconds. Given the heavy tensor calculations occurring in the cloud, this is considered a highly optimized and commercially viable latency metric.

Table 3 : Validation Results

Syste Component	Task / Process	Hardware / Environment	Performance Metric
Frontend (React UI)	Client-side rendering & animations	Local Browser	Constant 60 FPS
Local Orchestrator	Firebase Token Validation & Routing	Local CPU	< 50 milliseconds
Local Orchestrator	LLM Vocal Guide Generation	Google Gemini API	25 – 30 seconds
Remote GPU Engine	System Initialization (Cold Start)	Google Colab (NVIDIA T4)	~ 105 seconds (One-time)
Remote GPU Engine	Generative AI Synthesis (MusicGen)	Google Colab (NVIDIA T4)	80 – 90 seconds
Remote GPU Engine	DSP Pipeline (PyWorld + Pedalboard)	Google Colab (NVIDIA T4)	3 – 5 seconds
Overall Architecture	Total Round-Trip Time (RTT)	Distributed System	~ 120 seconds

6.4 Analysis of results

The final audio outputs generated by IsaiCraft were analyzed against standard industry mixing metrics to evaluate the effectiveness of the automated DSP pipeline.

Vocal Pitch & Timbre: The most significant result of the system is the efficacy of the "Humanized Soft-Pitch Correction." An acoustic analysis of the raw vocal input versus the processed vocal stem revealed that traditional aggressive autotune effectively flattens the frequency curve, destroying micro-fluctuations. In contrast, IsaiCraft's application of a 0.55 retune_strength successfully pulled flat and sharp notes into harmonic alignment with the generative beat while distinctly preserving the user's natural vibrato and emotional delivery. The resulting vocal sounded polished rather than artificial.

Dynamic Mastering: Analysis of the Master Mixbus output proved that the programmatic array normalization functioned flawlessly. By strictly clamping the generative instrumental peak to 0.35 and boosting the vocal stem to 0.90, the system prevented frequency masking. The vocals sat clearly "on top" of the mix. Furthermore, the application of the soft-limiter (`np.tanh`) successfully maximized the track's perceived RMS (Root Mean Square) loudness without introducing any digital clipping or distortion artifacts, yielding a track ready for immediate playback.

7. CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

The IsaiCraft project successfully validates the hypothesis that complex, studio-grade music production can be democratized for non-singers through the strategic application of distributed artificial intelligence and programmatic audio engineering. By decoupling the user interface from the heavy computational requirements of generative audio models, the system achieved a highly responsive, low-latency user experience on standard consumer hardware.

The successful implementation of the "Humanized Soft-Pitch Correction" algorithm using PyWorld stands as a critical advancement over traditional, rigid auto-tuning methods. By applying a fractional mathematical retune, the system proved capable of pulling off-key vocals into harmonic alignment with an AI-generated beat while explicitly preserving the emotional micro-fluctuations and natural vibrato of the user's voice. Furthermore, the automated Digital Signal Processing (DSP) Mixbus eliminated the need for manual equalization and compression, outputting a track that was dynamically balanced and immediately ready for playback. Ultimately, IsaiCraft successfully bridges the gap between passive AI music generation and authentic human participation, providing a highly secure, intuitive, and technically robust neuro-acoustic production studio.

7.2 Future Enhancements

While the current architecture successfully delivers end-to-end audio generation and mastering, several avenues exist for future technical expansion to elevate the platform to a full-scale commercial or enterprise application.

Currently, the system relies on browser localStorage and temporary Base64 encoding for session management, prioritizing speed and stateless security. A primary future enhancement will involve integrating a persistent NoSQL cloud database, such as Firebase Firestore or MongoDB. This will allow users to save multiple projects over time, re-download past stems (raw vocals, instrumental beats, and mastered tracks), and maintain a long-term portfolio of AI-assisted compositions.

At present, the Meta MusicGen model is prompted using text metadata (e.g., "Mood"). Future iterations will explore multimodal conditioning, specifically audio-to-audio prompting. By feeding the user's raw vocal melody directly into the transformer model as a conditioning tensor, the AI could generate an instrumental beat that not only matches the requested mood but also harmonically and rhythmically syncs to the specific cadence of the user's vocal performance.

To further expand the utility of the platform, future development will focus on implementing WebSocket connections (e.g., via Socket.io) to support real-time collaborative editing. This would allow multiple users across different geographic locations to simultaneously input lyrics, adjust generative prompts, and review vocal stems within a shared IsaiCraft studio session, effectively creating a multiplayer DAW environment.

Another significant enhancement lies in the cultural and regional fine-tuning of the generative models. Currently, foundational models like MusicGen are predominantly trained on Western music datasets, which can limit their ability to accurately synthesize complex regional genres, such as traditional Tamil folk or classical Carnatic music. Future development will focus on fine-tuning the transformer models using localized datasets encompassing regional instruments (e.g., Nadaswaram, Thavil) and complex microtonal scales (Ragas). Furthermore, the cognitive Large Language Model acting as the Vocal Director could be specialized to analyze Tamil poetic structures and meter (Yaappu), offering highly specific, culturally aware phonetic guidance tailored to regional lyricists.

Finally, to address long-term concerns regarding data privacy and cloud inference latency, future iterations of IsaiCraft will explore a transition toward a local-first desktop architecture. Utilizing modern framework wrappers such as Electron or Tauri, the application could be packaged as a standalone executable. By leveraging hardware-accelerated inferencing engines like TensorRT or ONNX Runtime, highly quantized versions of both the audio generation models and the DSP pipeline could be executed directly on the user's dedicated hardware. This architectural shift from a transient cloud environment to a local-first ecosystem would completely eliminate network bottlenecks, ensure absolute user data privacy by keeping all raw audio files strictly on-device, and provide a zero-latency production experience.

References

Research Papers and Journals

1. Copet, J., Kreuk, F., Gat, I., Remez, T., Kant, D., Synnaeve, G., Adi, Y., & Défossez, A. (2024). "Simple and Controllable Music Generation." Advances in Neural Information Processing Systems (NeurIPS). (Core research paper for Meta's MusicGen architecture).<https://arxiv.org/abs/2306.05284>
2. Gemini Team, Google DeepMind. (2023). "Gemini: A Family of Highly Capable Multimodal Models." arXiv preprint arXiv:2312.11805. (Technical architecture outlining the cognitive and text-processing capabilities of the Vocal Director) <https://arxiv.org/abs/2312.11805>
3. Ramirez, S. (2020). Building Data Science Applications with FastAPI. Packt Publishing. (Reference for establishing the asynchronous Python routing and API middleware) <https://www.packtpub.com/product/building-data-science-applications-with-fastapi/9781838821124>
4. Owsinski, B. (2017). The Mixing Engineer's Handbook (4th ed.). BMO Media. (Utilized as the theoretical basis for the programmatic EQ, Compression, and Reverb DSP chains).<https://bobbyowsinski.com/mixing-engineers-handbook/>
5. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. (Foundational textbook for understanding the tensor mathematics and transformer architectures utilized in the remote GPU engine)<https://www.deeplearningbook.org/>
6. McFee, B., Raffel, C., Liang, D., Ellis, D. P., McVicar, M., Battenberg, E., & Nieto, O. (2015). "librosa: Audio and music signal analysis in python." In Proceedings of the 14th Python in Science Conference (Vol. 8, pp. 18-25). <https://doi.org/10.25080/Majora-7b98e3ed-003>
7. Smus, B. (2013). Web Audio API: Advanced Sound for Games and Interactive Apps. O'Reilly Media, Inc. (Reference for frontend raw microphone audio acquisition and blob encoding). <https://webaudioapi.com/>
8. Smith, J. O. (2007). Introduction to Digital Filters with Audio Applications. W3K Publishing. (Reference material for Highpass, Lowpass, and Peak filtering logic in audio arrays). <https://ccrma.stanford.edu/~jos/filters/>
9. Boll, S. (1979). "Suppression of acoustic noise in speech using spectral subtraction." IEEE Transactions on Acoustics, Speech, and Signal Processing, 27(2), 113-120. (Foundational acoustic mathematics utilized by the noise reduction modules).

Technical Documentation and Web Resources

1. Google Developers. (2026). Firebase Authentication and Admin SDK Official Documentation. <https://firebase.google.com/docs/auth>
2. Meta / React Core Team. (2026). React: The library for web and native user interfaces. <https://react.dev/>
3. Mozilla Developer Network (MDN). (2026). Web Audio API Documentation. https://developer.mozilla.org/en-US/docs/Web/API/Web_Audio_API
4. Ngrok. (2026). Ngrok Secure Tunnels Documentation. <https://ngrok.com/docs/>
5. Tiangolo. (2026). FastAPI Documentation: Web framework for building APIs with Python 3.8+. <https://fastapi.tiangolo.com/>